Roll No: _____

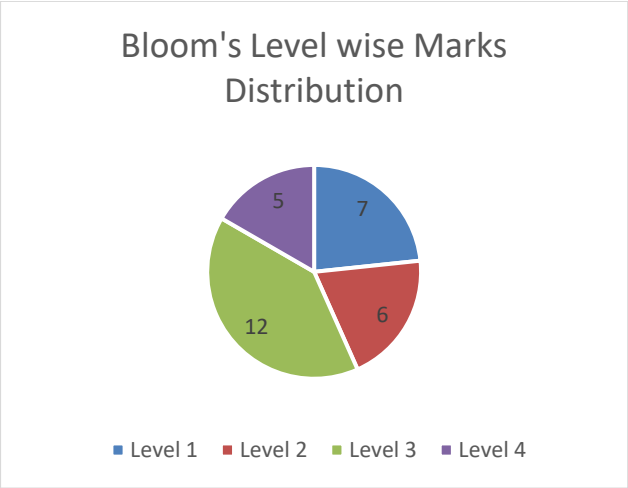
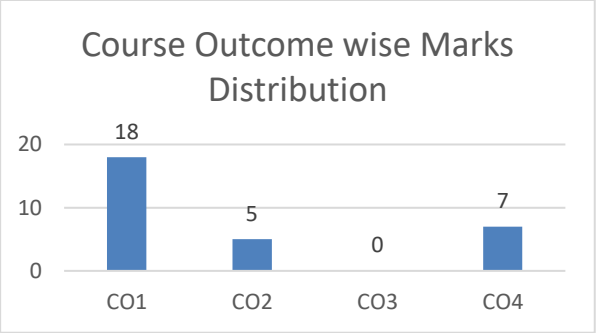
Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
MID Semester Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 2 hours (11:00 am-1:00 pm)
Marks : 30

Course Code: UCS619
Date: March 16, 2024
Name of Faculty: Dr. Ajay Kumar, Dr Samya Muhuri

Note: Attempt all questions. Assume missing data, if any suitably. Give appropriate reasoning for your answers.

Q.No	Questions	Marks	CO	BL
1.a.	A quantum state $\|\phi\rangle$ is given as $$\|\phi\rangle = \frac{i}{2}\|0\rangle + \frac{\sqrt{3}}{2}\|1\rangle$$ Show the state is normalized or not.	1	CO1	L3
1.b.	Compute $\langle\phi\|$.	1	CO1	L3
1.c.	Indicate the probability of finding $\|0\rangle$ and $\|1\rangle$ when a measurement is made.	1	CO1	L3
1.d.	Compute the density matrix(ρ) of ϕ.	1	CO1	L3
1.e.	Measure the $\|\phi\rangle$ in Hadamard basis.	2	CO1	L3
2.a.	Consider two-qubit quantum state $$\|\phi\rangle = \frac{\sqrt{2}}{3\sqrt{3}}\|00\rangle - \frac{1}{\sqrt{6}}\|01\rangle + \frac{2i\sqrt{2}}{3\sqrt{3}}\|10\rangle - \frac{5i}{3\sqrt{6}}\|11\rangle$$ What are the probabilities of outcome of 0 and 1 if first qubit of $\|\phi\rangle$ is measured?	2	CO1	L3
2.b.	What are the probabilities of outcome 0 and 1 if second qubit of $\|\phi\rangle$ is measured?	2	CO1	L3
2.c.	What is the state of the system after first qubit is measured as $\|1\rangle$?	2	CO1	L3
3.a.	Using standard quantum gates, construct a 2-qubit unitary circuit with an unitary operator U, that copies the state $\|+\rangle$. The function of the operator is as follows: $U(\|+\rangle\|0\rangle) = \|+\rangle\|+\rangle$	2	CO2	L4
3.b.	Create a circuit for $(x_0 \vee x_1) \wedge (x_1 \vee x_2)$. Here $\wedge$ and $\vee$ represent AND and OR Gate, respectively. x_0, x_1 and x_2 represents first, second and third qubits. You can use only three additional qubits. Draw the quantum circuit.	3	CO2	L4
4.a.	Write down the different steps for creating shifted key based on E91 protocol by considering the entangled pair $\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$. Explain it with the help of an example.	4	CO4	L1

4.b.	Explain the procedure of Super dense Coding assuming that the third party generates the entangled pair $\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$. You will not allowed to consider the starting state as $\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$.	3	CO4	L2
5.a.	Define complex vector space. Explain it with the help of an example.	3	CO1	L1
5.b.	List down various properties of Hilbert space. Explain with the help of examples.	3	CO1	L2

Marks Distribution